

Using Limit Comparison Test to show $\sum_{n=1}^{\infty} a_n$ converges:

Step 1: Form a new series $\sum_{n=1}^{\infty} b_n$ similar to $\sum_{n=1}^{\infty} a_n$;

Step 2: Show that $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L > 0$; and show that $\sum_{n=1}^{\infty} b_n$ converges.

Using Limit Comparison Test to show $\sum_{n=1}^{\infty} a_n$ diverges:

Step 1: Form a new series $\sum_{n=1}^{\infty} b_n$ similar to $\sum_{n=1}^{\infty} a_n$;

Step 2: Show that $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L > 0$; and show that $\sum_{n=1}^{\infty} b_n$ diverges.

1) Show that the following series converges: $\sum_{n=1}^{\infty} \frac{2n^2 - 1}{7n^5 + 2n + 1}$

Hint: Let $a_n = \frac{2n^2 - 1}{7n^5 + 2n + 1}$ Let $b_n = \frac{n^2}{n^5} = \frac{1}{n^3}$

a) $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{\frac{2n^2 - 1}{7n^5 + 2n + 1}}{\frac{1}{n^3}} = \lim_{n \rightarrow \infty} \frac{2n^5 - 1}{7n^5 + 2n + 1} = \underline{\hspace{1cm}} ?$ Hint: Use L'Hopital's Rule

b) Explain why the series $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^3}$ converges.

c) $\sum_{n=1}^{\infty} \frac{1}{n^3} = \underline{\hspace{1cm}}$ (finite value or ∞ ?)

d) Use the Limit Comparison Test to explain why the series $\sum_{n=1}^{\infty} \frac{2n^2 - 1}{7n^5 + 2n + 1}$ converges.

2) Show that the following series converges: $\sum_{n=1}^{\infty} \frac{1}{n\sqrt{n^2+5}}$

Hint: Let $a_n = \frac{1}{n\sqrt{n^2+5}}$ Let $b_n = \frac{1}{n\sqrt{n^2}} = \frac{1}{n^2}$

$$\text{a) } \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{\frac{1}{n\sqrt{n^2+5}}}{\frac{1}{n\sqrt{n^2}}} = \lim_{n \rightarrow \infty} \frac{\sqrt{n^2}}{\sqrt{n^2+5}} = \lim_{n \rightarrow \infty} \sqrt{\frac{n^2}{n^2+5}} = \sqrt{\lim_{n \rightarrow \infty} \frac{n^2}{n^2+5}} = \underline{\hspace{2cm}} ?$$

b) Explain why the series $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.

$$\text{c) } \sum_{n=1}^{\infty} \frac{1}{n^2} = \underline{\hspace{2cm}} \text{ (finite value or } \infty \text{?)}$$

d) Use the Limit Comparison Test to explain why the series $\sum_{n=1}^{\infty} \frac{1}{n\sqrt{n^2+1}}$ converges.

3) Show that the following series converges: $\sum_{n=1}^{\infty} \frac{2n^2 + 3n}{3^n(n^2 + 2)}$

Hint: Let $a_n = \frac{2n^2 + 3n}{3^n(n^2 + 2)}$ and $b_n = \frac{1}{3^n}$

a) $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} =$ _____

b) Use the Geometric Series Test to explain why the $\sum_{n=1}^{\infty} \frac{1}{3^n}$ converges.

c) Use the Limit Comparison Test to explain why the $\sum_{n=1}^{\infty} \frac{2n^2 + 3n}{3^n(n^2 + 2)}$ converges.

4) Show that the following series converges: $\sum_{n=1}^{\infty} \frac{2n + \sqrt{n}}{n^{5/2} + 7n + 3}$

Hint: Let $a_n = \frac{2n + \sqrt{n}}{n^{5/2} + 7n + 3}$ and $b_n = \frac{n}{n^{5/2}} = \frac{1}{n^{3/2}}$

a) $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} =$ _____

b) Use the p -series Test to explain why the $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ converges.

c) Use the Limit Comparison Test to explain why the $\sum_{n=1}^{\infty} \frac{2n + \sqrt{n}}{n^{5/2} + 7n + 3}$ converges.

5) Show that the following series converges: $\sum_{n=1}^{\infty} \frac{1}{n \cdot \sqrt[3]{n+3}}$

Hint: Let $a_n = \frac{1}{n \cdot \sqrt[3]{n+3}}$ and $b_n = \frac{1}{n \cdot \sqrt[3]{n}}$ and

a) Find $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} =$ _____

b) Explain why the series $\sum_{n=1}^{\infty} \frac{1}{n \cdot \sqrt[3]{n}}$ converges.

c) Use the Limit Comparison Test to explain why the series $\sum_{n=1}^{\infty} \frac{1}{n \cdot \sqrt[3]{n+3}}$ converges.

6) Show that the following series converges: $\sum_{n=1}^{\infty} \frac{\ln n}{n^3}$

Hint: Let $a_n = \frac{\ln n}{n^3}$ and $b_n = \frac{n}{n^3}$

a) $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} =$ _____

b) Use the p -series Test to explain why the $\sum_{n=1}^{\infty} \frac{n}{n^3}$ converges.

c) Use the Limit Comparison Test to explain why the $\sum_{n=1}^{\infty} \frac{\ln n}{n^3}$ converges.

Hint: Use Case 2 of the Limit Comparison Test Proof (see proof above).

Case 2: Claim: If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ and $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges, where $a_n > 0$ and $b_n > 0$.

7) Show that the following series converges: $\sum_{n=1}^{\infty} \frac{n + \ln n}{n^4 + 1}$

Hint: $\sum_{n=1}^{\infty} \frac{n + \ln n}{n^4 + 1} = \sum_{n=1}^{\infty} \frac{n}{n^4 + 1} + \sum_{n=1}^{\infty} \frac{\ln n}{n^4 + 1}$

a) Explain why $\sum_{n=1}^{\infty} \frac{n}{n^4 + 1}$ converges.

b) Explain why $\sum_{n=1}^{\infty} \frac{\ln n}{n^4 + 1}$ converges.

8) Show that the following series diverges: $\sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{4n^2+1}}$

Hint: Let $a_n = \frac{1}{\sqrt[3]{4n^2+1}}$ and $b_n = \frac{1}{\sqrt[3]{n^2}} = \frac{1}{n^{2/3}}$

a) Explain why $\sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n^2}}$ diverges.

b) Explain why $\sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{4n^2+1}}$ diverges.

